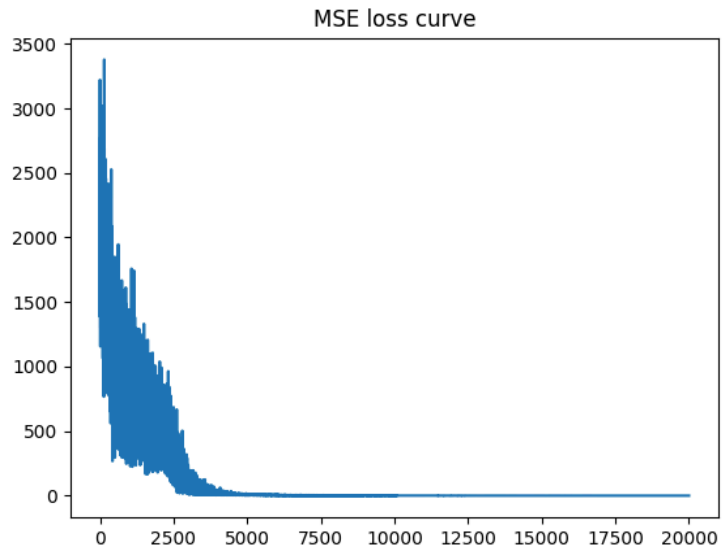
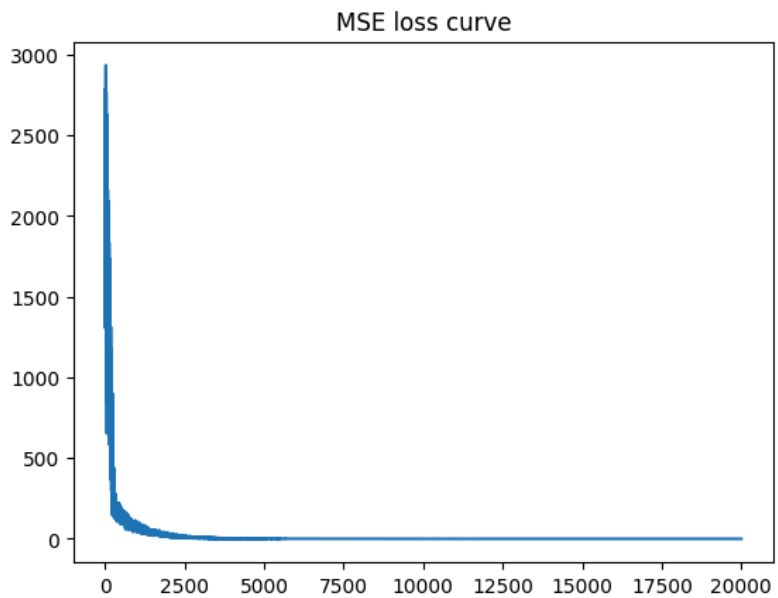
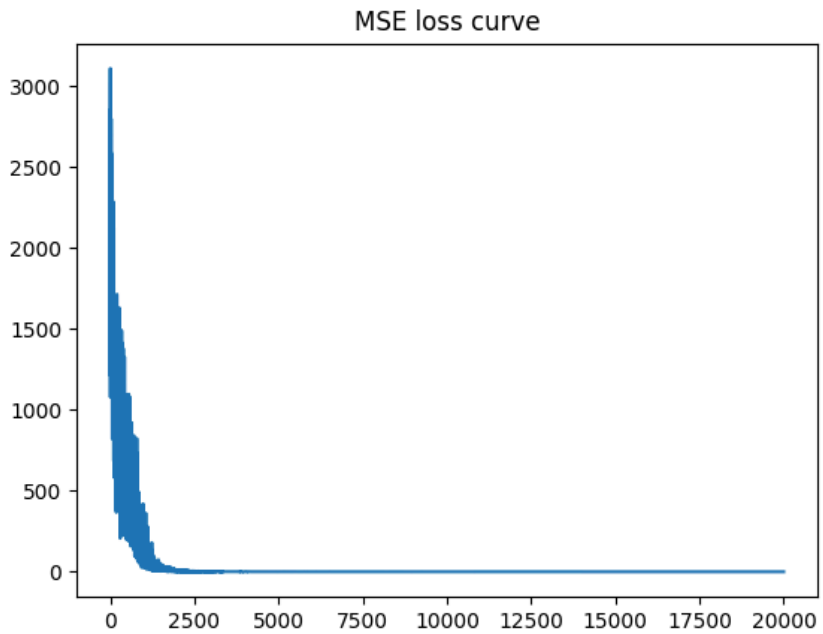
Q1.)
Plot: 16 layers



64 layers



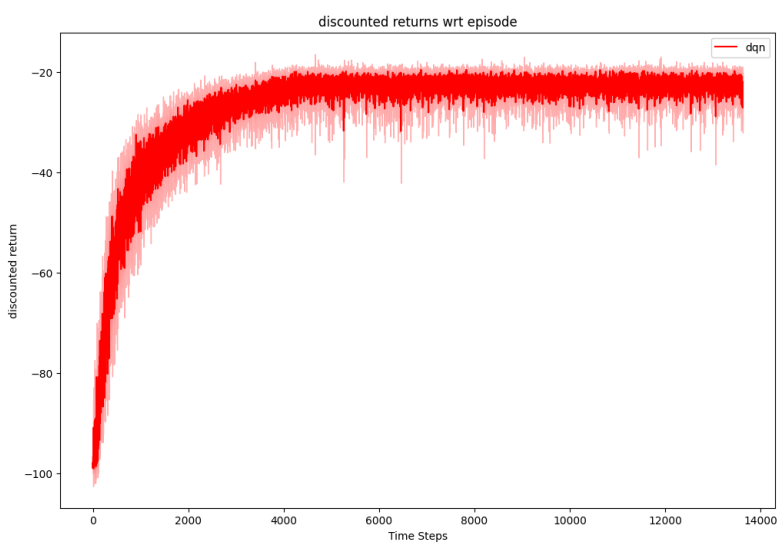
128 layers

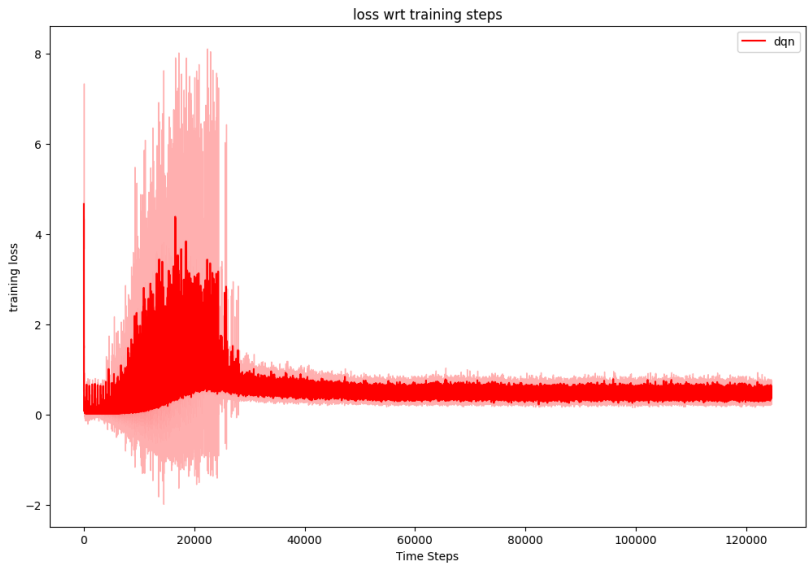


e.) How accurate is your learned model? How accurate is it within the range of [−10,10]? How about outside of the range? Can you notice any difference between models having different layer widths? What do you think may have caused the difference?

It was pretty accurate. It is within the range of -10 to 10. It starts by shooting outisde of that range but then it converges. I noticed that at 64 layers it learns the best. With more or less I either don't learn enough or I learn too much and my policy overshoots. The difference is how much information I am feeding into my policy.

Q2.)
Plot:

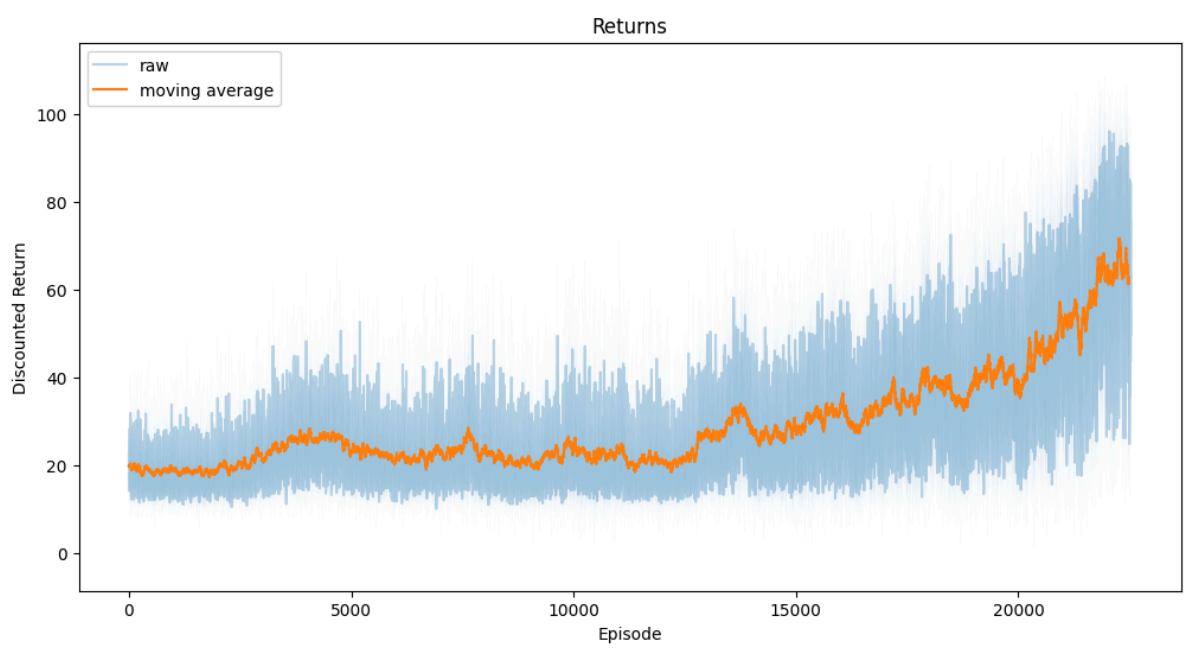


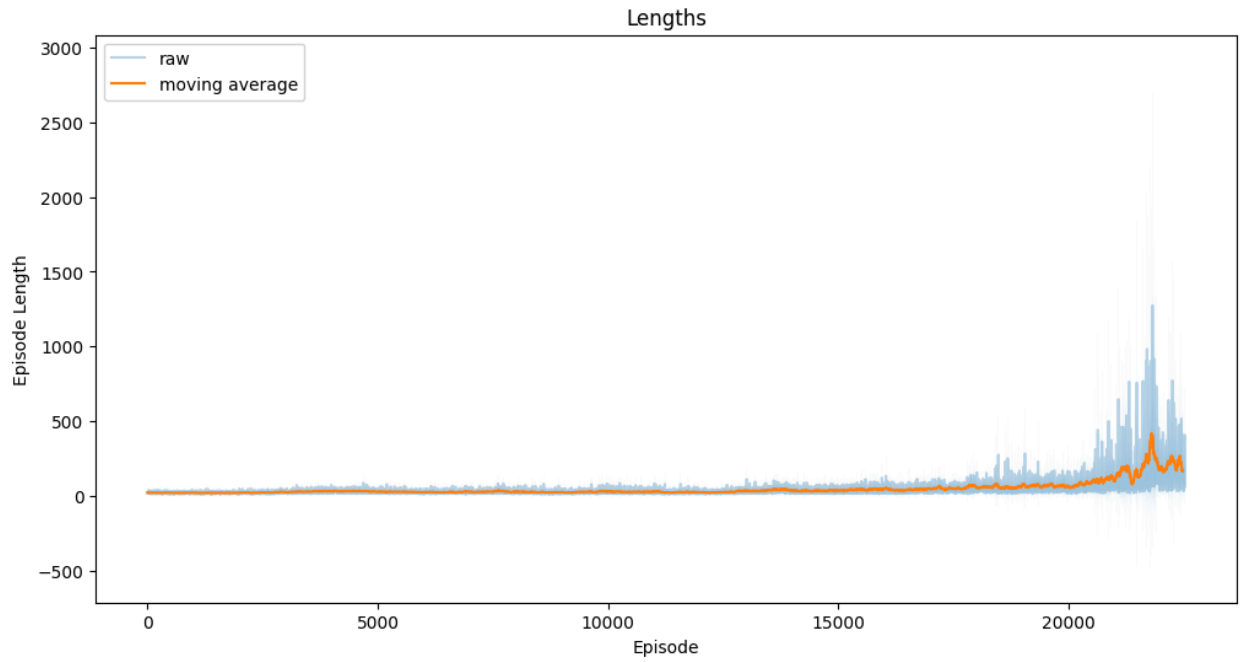


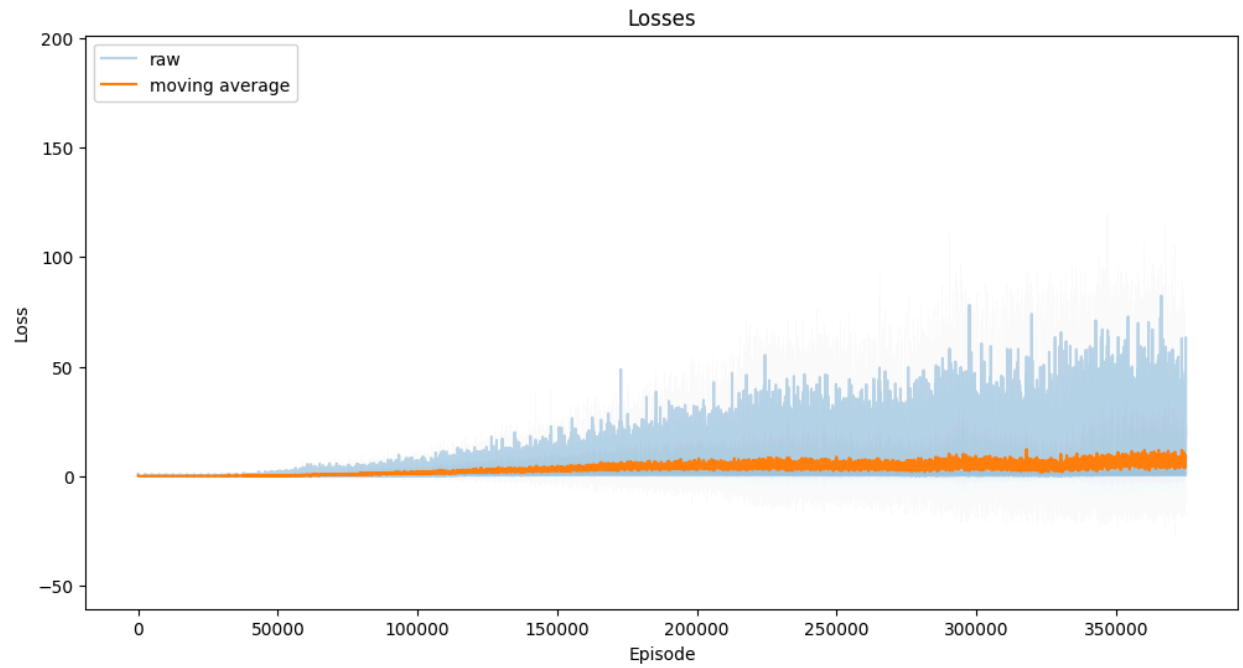
d.) How does it compare to tabular methods from previous exercises?

It has a very steep learning rate. This coincides with the large loss learning that happens at the beginning of the training. Over all it trains very well compared to previous exercises, and doesn't take long to be performing well, but it does take a large amount of episodes in order to learn well, so it's not the fastest/

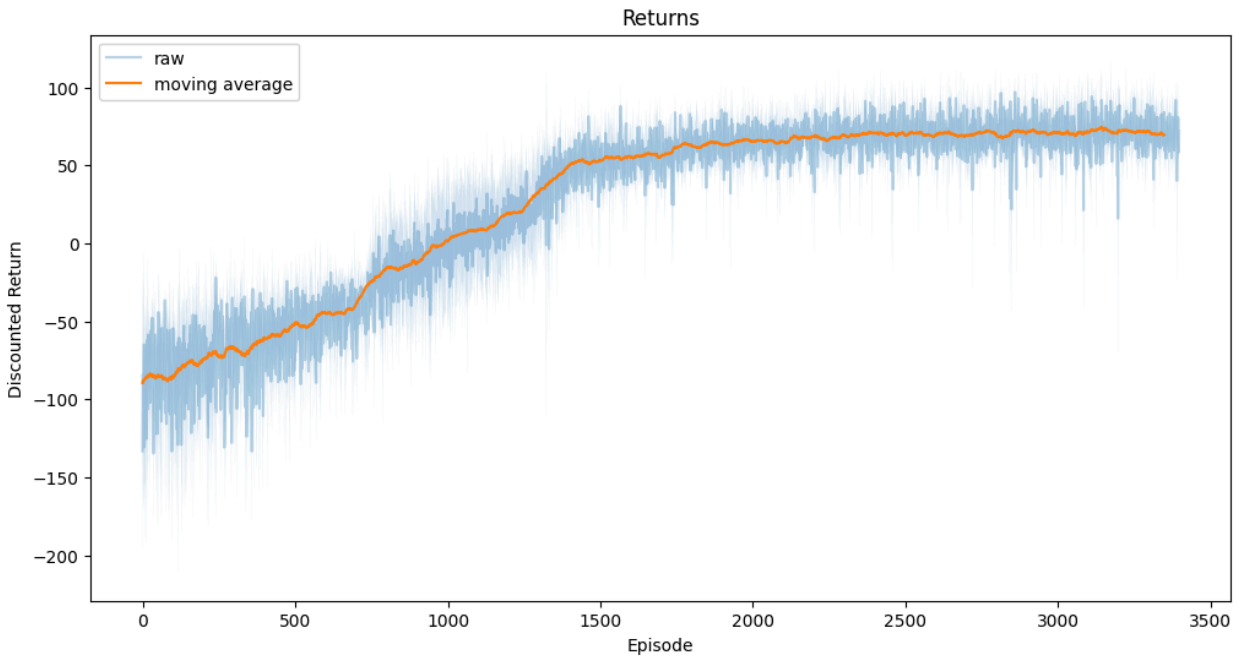
Q3.)
Plot: CartPole







Plot: LunarLander


Returns
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moving average
Discounted Return
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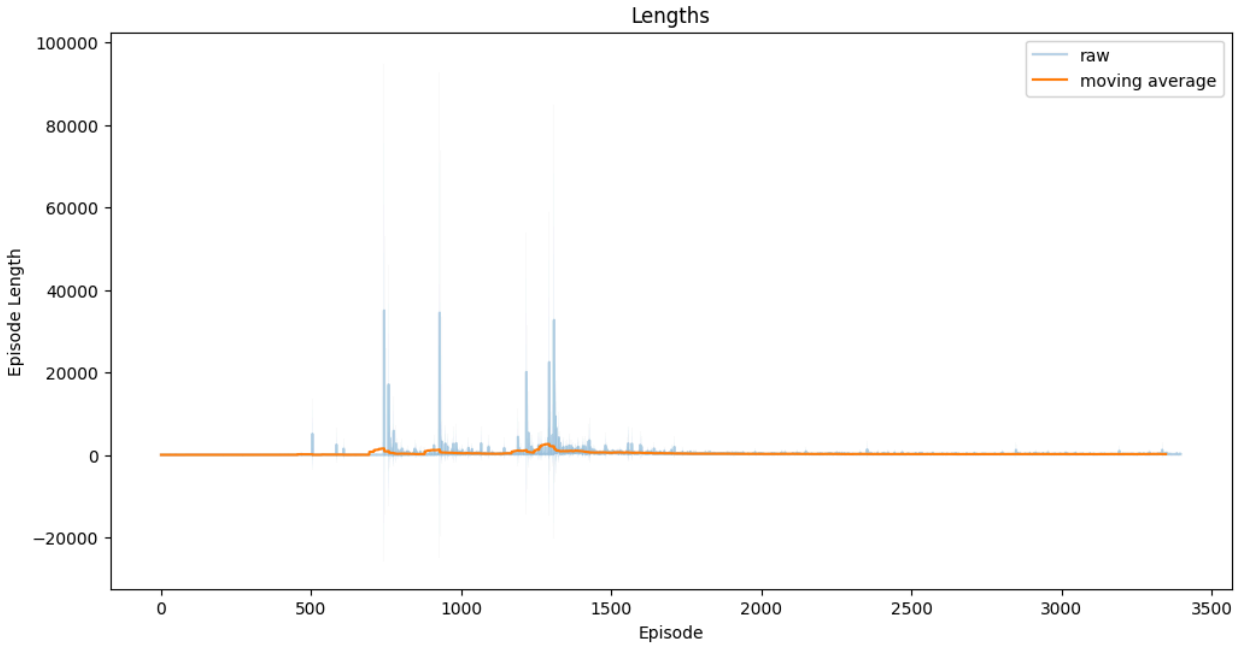
50
0
−50
−100
−150
−200
0
500
1000
1500
2000
2500
3000
3500
Episode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moving average
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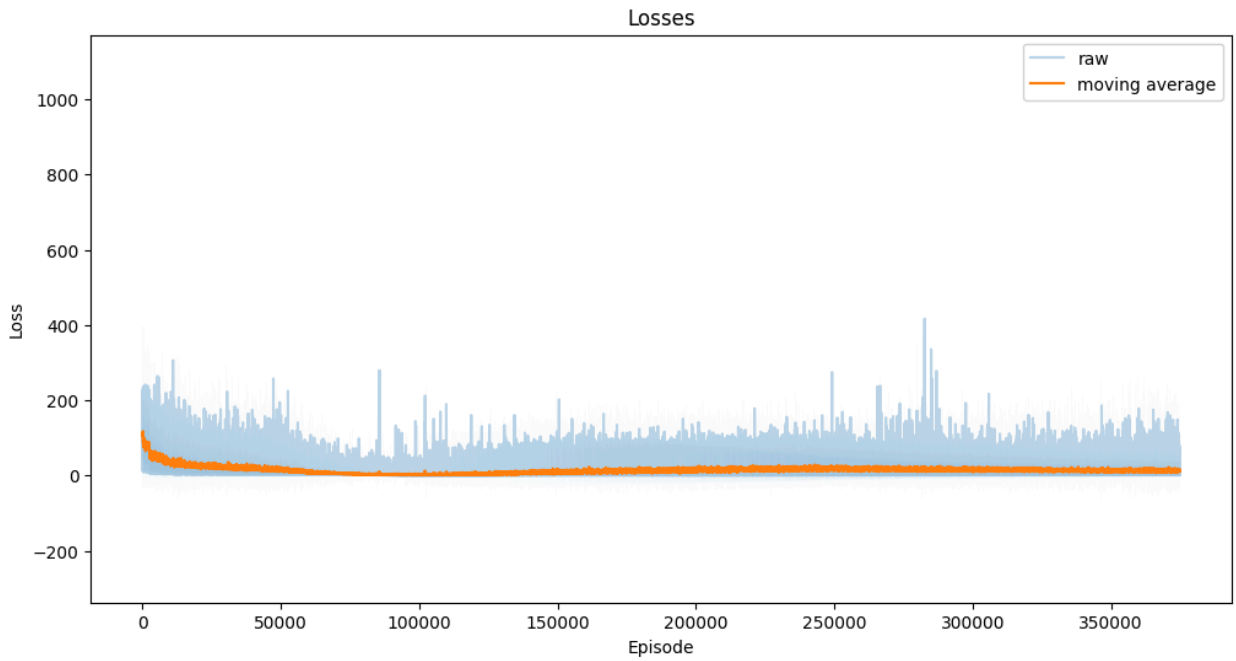
Episode



d.) Render your runs at different stages of training, what kind of policy (or policies) do you see? What do you think the corresponding value function would be? You can inspect your value function to validate your intuition.

My answers in the notebook. I think the corresponding value functions would be roughly uniform and love everywhere, then it would start to have some differentiation around the midpoint, and then towards the end, the value function would be well shaped.